BetEdge Analytics - NBA Game Prediction Model

Problem Statement: How can BetEdge build a machine learning model that predicts NBA game outcomes with a 60% accuracy rate, in order to provide clients with data-driven insights for game predictions?

Context: BetEdge Analytics is a sports analytics startup that provides NBA game prediction insights to its clients. The company currently employs analysts who manually review box scores and team statistics to predict game outcomes achieving roughly 52% accuracy, barely above random guessing.With 1,230 regular season NBA games per season, manually analyzing every matchup is time consuming and inconsistent. The company wants to replace this manual process with a machine learning model that can automatically analyze game data and predict outcomes more accurately than their current analysts. They hope to achieve a model that can predict 60% accuracy as soon as possible so they could make a big advertisement before the season starts.

Criteria for Success: Model achieves 60%+ prediction accuracy, Identifies which features most influence game outcomes, and can process new game data automatically

Scope of solution: Focused exclusively on NBA regular season games, use team level performance metrics, and last 3 seasons of game data

Constraints : Only have access to data that's available to the public. Teams change every year. Injuries happen all the time and are hard for predictions. We have a sample size limited to the last three years due to relevance and teams changing.

Stakeholders: Marcus Webb - CEO and cofounder, Derek Chen - Head of Data and Technology James Okafor - Senior Analyst.

Data Sources :NBA Stats API (stats.nba.com) - contains official game by game statistics and team performance metrics

Basketball Reference (basketball-reference.com) - contains historical game data and advanced team statistics

Kaggle NBA datasets - contains multiple seasons of historical NBA data

Deliver